Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-6(P1) | Date: 16-04-26
Time: 3 Hrs | 2022_P1 | Max. Marks: 180

KEY SHEET
MATHS

1	**8**	2	**6**	3	**1**	4	**0.50**	5	**2.50**
6	**4**	7	**43**	8	**2**	9	**ABD**	10	**ABCD**
11	**AB**	12	**AC**	13	**AD**	14	**ABD**	15	**A**
16	**B**	17	**B**	18	**D**				

PHYSICS

19	**4**	20	**1**	21	**1.15 TO 1.25**	22	**4**	23	**1**
24	**6**	25	**375**	26	**2**	27	**CD**	28	**ABC**
29	**BCD**	30	**ABC**	31	**ABC**	32	**ABD**	33	**A**
34	**C**	35	**D**	36	**D**				

CHEMISTRY

37	**88.89**	38	**6.25**	39	**0.56 TO 0.58**	40	**39**	41	**17**
42	**1**	43	**0.65**	44	**10.25**	45	**ABCD**	46	**ABC**
47	**BCD**	48	**ACD**	49	**ACD**	50	**ABD**	51	**D**
52	**C**	53	**C**	54	**C**				

For More Join: @IITJEE_Advanced

SOLUTIONS
MATHS

1. $M=0,\ m=8$
2. Roots are $-1,2,3,\&7$
3. $(1,1),(1,2),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)$
5. $t_n=\dfrac{n+2}{n(n+1)(n+3)}$

$=\dfrac{1}{6}\left[\dfrac{1}{n+1}-\dfrac{1}{n+2}\right]+\dfrac{1}{6}\left[\dfrac{1}{n+2}-\dfrac{1}{n+3}\right]+\dfrac{2}{3}\left[\dfrac{1}{n}-\dfrac{1}{n+1}\right]$

So, $S=\sum_{n=1}^{n}T_n=\dfrac{1}{6}\sum_{n=1}^{n}\left(\dfrac{1}{n+1}-\dfrac{1}{n+2}\right)+\dfrac{1}{6}\sum_{n=1}^{n}\left(\dfrac{1}{n+2}-\dfrac{1}{n+3}\right)+\dfrac{2}{3}\sum_{n=1}^{n}\left(\dfrac{1}{n}-\dfrac{1}{n+1}\right).$

$=\dfrac{1}{6}\left(\dfrac{1}{2}-\dfrac{1}{n+1}\right)+\dfrac{1}{6}\left(\dfrac{1}{3}-\dfrac{1}{n+1}\right)+\dfrac{2}{3}\left(1-\dfrac{1}{n+1}\right)$

$=\dfrac{1}{6}\left[\dfrac{29}{6}-\dfrac{4}{n+1}-\dfrac{1}{n+2}-\dfrac{1}{n+3}\right]$ (proved)

6. $m=2(-2\ \&\ 4,2,10$ are not in the domain)
7. $A=5,\ b=8,c=9\ \&\ d=11$
8. STATEMENT 1 :

Let $\overrightarrow{OA}=\hat{a},\overrightarrow{OB}=\hat{b},\overrightarrow{OC}=\hat{c}$ and $\overrightarrow{OP}=\hat{d}$

Here, $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\dfrac{1}{2}$

And $\hat{d}\cdot\hat{a}=\hat{d}\cdot\hat{b}=\hat{d}\cdot\hat{c}=\cos\alpha$

Let, $\hat{d}=x\hat{a}+y\hat{b}+z\hat{c}$

$\hat{d}\cdot\hat{a}=x+\dfrac{y}{2}+\dfrac{z}{2}=\cos\alpha\Rightarrow\hat{d}\cdot\hat{b}=\dfrac{x}{2}+y+\dfrac{z}{2}=\cos\alpha\Rightarrow\hat{d}\cdot\hat{c}=\dfrac{x}{2}+\dfrac{y}{2}+z=\cos\alpha$

Adding we get $2(x+y+z)=3\cos\alpha$

Taking dot product with $\hat{d}$ in (1) we get

$1=x\cos\alpha+y\cos\alpha+z\cos\alpha\Rightarrow x+y+z=\dfrac{1}{\cos\alpha}$

$\Rightarrow\dfrac{3\cos\alpha}{2}=\dfrac{1}{\cos\alpha}\Rightarrow\cos^2\alpha=\dfrac{2}{3}\Rightarrow 3p_1=3\cos^2\alpha=2$

STATEMENT 2 :

Note that $\overrightarrow{AB}^2+\overrightarrow{CD}^2=3^2+11^2=130=7^2+9^2=\overrightarrow{BC}^2+\overrightarrow{DA}^2$

Since $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}+\overrightarrow{DA}=\vec{0}$,

$DA^2=(\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD})^2=AB^2+BC^2+CD^2+2\left(\overrightarrow{AB}\cdot\overrightarrow{BC}+\overrightarrow{BC}\cdot\overrightarrow{CD}+\overrightarrow{CD}\cdot\overrightarrow{AB}\right)$

$=AB^2-BC^2+CD^2+2\left(\overrightarrow{BC}^2+\overrightarrow{AB}\cdot\overrightarrow{BC}+\overrightarrow{BC}\cdot\overrightarrow{CD}+\overrightarrow{CD}\cdot\overrightarrow{AB}\right)$

$=AB^2-BC^2+CD^2+2\left(\overrightarrow{AB}+\overrightarrow{BC}\right)\cdot\left(\overrightarrow{BC}+\overrightarrow{CD}\right),$

i.e., $2\overrightarrow{AC}\cdot\overrightarrow{BD}=2\left(\overrightarrow{AB}+\overrightarrow{BC}\right)\cdot\left(\overrightarrow{BC}+\overrightarrow{CD}\right)=AD^2+BC^2-AB^2-CD^2=0$

Hence both AC and BD are perpendicular to each other

$\Rightarrow p_2=\cos\theta=0$

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9. $f(x)=x^2+5x-4$

10. $f(x)=\frac{x}{2}\left(e^{\frac{x^2}{2}+\frac{x}{6}}\right)$

11. A) $x_n=\frac{1}{2}\left(x_{n-1}+\frac{5}{x_{n-1}}\right)=\frac{\sqrt{5}}{2}\left(\frac{x_{n-1}}{\sqrt{5}}+\frac{\sqrt{5}}{x_{n-1}}\right)$

$\Rightarrow \frac{x_n}{\sqrt{5}}=\frac{1}{2}\left(\frac{x_{n-1}}{\sqrt{5}}+\frac{5}{x_{n-1}}\right)$.

By componendo-dividend, we get

$$\Rightarrow \frac{x_n-\sqrt{5}}{x_n+\sqrt{5}}=\frac{\frac{1}{2}\left(\frac{x_{n-1}}{\sqrt{5}}+\frac{\sqrt{5}}{x_{n-1}}\right)-1}{\frac{1}{2}\left(\frac{x_{n-1}}{\sqrt{5}}+\sqrt{\frac{5}{x_{n-1}}}\right)+1}=\frac{x_{n-1}^2+\sqrt{5}-2\sqrt{5}x_{n-1}}{x_{n-1}^2+\sqrt{5}+2\sqrt{5}x_{n-1}}$$

$$\Rightarrow \left(\frac{x_{n-1}-\sqrt{5}}{x_{n-1}+\sqrt{5}}\right)^2=\left(\frac{x_{n-2}-\sqrt{5}}{x_{n-2}+\sqrt{5}}\right)^{2^2}=\ldots\ldots=\left(\frac{x_1-\sqrt{5}}{x_1+\sqrt{5}}\right)^{2n-1}$$

$$=\left(\frac{x_1-\sqrt{5}}{x_1+\sqrt{5}}\right)^{2n} \text{ (proved)}$$

B) Since x is a positive number, $x-\sqrt{5}<x+\sqrt{5}$

$$\Rightarrow \frac{x-\sqrt{5}}{x+\sqrt{5}}<1$$

$$\therefore \left(\frac{x-\sqrt{5}}{x+\sqrt{5}}\right)^{2n}\to 0 \text{ as } n\to\infty.$$

$$\Rightarrow \frac{x-\sqrt{5}}{x+\sqrt{5}}\to 0\Rightarrow x-\sqrt{5}\to 0$$

$\therefore \text{Lt}_{n\to\infty}x_n=\sqrt{5}$.

13. $$P=\begin{pmatrix}1&0&0\\0&1&1\\0&1&0\end{pmatrix},\begin{pmatrix}1&0&1\\0&1&0\\1&0&0\end{pmatrix},\begin{pmatrix}1&0&0\\0&0&1\\0&1&1\end{pmatrix},\begin{pmatrix}1&1&0\\1&0&0\\0&0&1\end{pmatrix},\begin{pmatrix}0&1&0\\1&1&0\\0&0&1\end{pmatrix},\begin{pmatrix}0&0&1\\0&1&0\\1&0&1\end{pmatrix},$$

14. Clearly, $\vec{a}=AB+AC+AD$

Also it is given that, $\overrightarrow{AB}\times\overrightarrow{AC}=\vec{b}$ and $\overrightarrow{AB}\times\overrightarrow{AD}=\vec{c}$

$\Rightarrow\left(\overrightarrow{AB}\times\overrightarrow{AC}\right)\times\vec{a}=\vec{b}\times\vec{a}$ and $\left(\overrightarrow{AB}\times\overrightarrow{AD}\right)\times\vec{a}=\vec{c}\times\vec{a}$

Also given $\frac{\overrightarrow{AB}\cdot\vec{a}}{|\vec{a}|}=1,\frac{\overrightarrow{AC}\cdot\vec{a}}{|\vec{a}|}=1\Rightarrow\overrightarrow{AB}\cdot\vec{a}=3,\overrightarrow{AC}\cdot\vec{a}=3$

Solving (1) and (2) and (3) we get desired result

15. $f(x)=0$

$g(x)=\pi$

$$h(x)=\begin{cases}0, x\in[0,1)\\ \pi, x\in[1,\infty)\end{cases}$$

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16. a) $\left\{\frac{3\pi}{10}, n \in z\right\}$ b) $\left\{\frac{3\pi}{3}+\frac{\pi}{6}, \frac{2\pi}{5}+\frac{\pi}{10}, n \in z\right\}$

c) $\left\{\pi n+\frac{n\pi}{5}+\frac{\pi}{10}, n \in z\right\}$ d) $\left\{\pi n+\frac{n\pi}{3}+\frac{\pi}{6}, n \in z\right\}$

17.

$\log_{10}(2xy) = 4+(\log_{10}x-1)(\log_{10}y-2)$

$[\log_{10}(2yz) = 4+(\log_{10}y-2)(\log_{10}z-1)\}$

$\log_{10}(zx) = 2+(\log_{10}z-1)(\log_{10}x-1)$

Let $\log x-1 = A$

$\log y-2 = B$

$\log_{10}z-1 = C$

In 'S'

$$\begin{cases} 4+AB = \log 2+A+1+B+2 \Rightarrow (A-1)(B-1) = \log 2 \\ 4+BC = \log 2+B+2+C+1 \Rightarrow (B-1)(C-1) = \log 2 \\ 2+AC = C+1+A+1 \Rightarrow (A-1)(C-1) = 1 \end{cases}$$

$\Rightarrow (A-1)(B-1)(C-1) = \pm\log 2$

$C-1 = \pm 1$

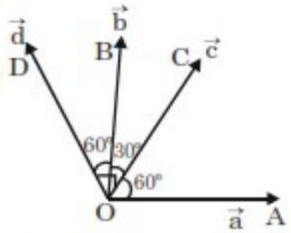
$A-1 = \pm 1$

$B-1 = \pm\log 2$

$(A,B,C) \equiv (2,1+\log 2,2),(0,1-\log 2,0)$

$(x,y,z) \equiv (10^3,2000,10^3),(10,500,10)$

18.



($\hat{p}$ is unit vector $\perp$ to the plane)

$\sin\theta_1\sin\theta_2\cos\phi = 1, \cos\alpha = \frac{1}{2}$

$\theta_1 = \theta_2 = 90^\circ, \phi = 0, \alpha = \frac{\pi}{3}$

$\Rightarrow$ OABCD are coplanar

$\Delta ABC = \frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$

$= \frac{1}{2}\left|2\hat{p}-3\hat{p}-\frac{3\sqrt{3}}{2}\hat{p}\right| = \frac{1}{2}\left(1+\frac{3\sqrt{3}}{2}\right)$

$\Delta BCD = \frac{1}{2}\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{d}+\vec{d}\times\vec{b}\right|$

$= \frac{1}{2}\left|-3\hat{p}+12\hat{p}-4\sqrt{3}\hat{p}\right| = \frac{1}{2}\left(9-4\sqrt{3}\right)$

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PHYSICS

19. $l = 94 \pm 0.1$ cm

$r = 5 \pm 0.05$ cm

$f = \frac{v}{4L}, L = l + 1.2r$

$\frac{\Delta f}{f} \times 100 = \frac{\Delta L}{L} \times 100$

$\alpha^2 = \frac{\Delta l + 1.2 \times \Delta r}{l + 1.2 \times r} \times 100$

$= 0.16$

$\alpha = 0.4, 10\alpha = 4$

20. $mg\cos\theta + kx = \frac{mv^2}{R}$ ____ (1)

$mg\frac{R}{2} = \frac{1}{2}mv^2 ----(2)$

$mv^2 = mg$

$kx = \frac{mg}{2}$

$x = \frac{2 \times 10}{2 \times 20} = 0.5$

$l_0 + x = R$

$l_0 = 1m$

21. $V_0 = \sqrt{\frac{GM}{R}}$

The first impulse point is apogee (R) of the elliptical path perigee must be 2R.

Velocity of apogee $= v_0 + \Delta v_1$

Velocity of perigee $= \frac{v_0 + \Delta v_1}{2}$

$\frac{v_0 + \Delta v_1}{2} + \Delta v_2 = \sqrt{2}v_0$

Also, $\frac{-GMm}{3R} = \frac{-GMm}{R} + \frac{1}{2}m\left(v_0 + \Delta v_1\right)^2$

$\frac{2GMm}{3R} = \frac{1}{2}m\left(v_0 + \Delta v_1\right)^2$

$\sqrt{\frac{4GM}{3R}} = \left(v_0 + \Delta v_1\right)$

$\frac{2}{\sqrt{3}}v_0 = v_0 + \Delta v_1$

$\Delta v_1 = \left(\frac{2}{\sqrt{3}} - 1\right)v_0$

$\Delta v_2 = \frac{1}{\sqrt{2}}v_0 - \frac{1}{\sqrt{3}}v_0\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}\right)v_0$

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22. $\left[L^3T^{-1}\right]=\left[ML^{-1}T^{-1}\right]^a[L]^b\left[ML^{-2}T^{-2}\right]^c$

$\Rightarrow a+c=0$

$-a+b-2c=3$

$\Rightarrow a+b=3$

and $-a-2c=-1$

$\Rightarrow c=1, a=-1, b=4$

$a+b+c=4$

23. $V_0=1cm^3$

$\gamma=1.8\times10^{-4}K^{-1}$

$A=1.8\times10^{-4}\ \text{cm}^2$

$\Delta V=V_0\gamma\Delta T=A\times\Delta h;\Delta T=1K$

$\Delta h=\dfrac{1\times1.8\times10^{-4}}{1.8\times10^{-4}}\ \text{cm}=1\ \text{cm}$

24. $x_1+x_2=x; x=\dfrac{27}{4}mm, x_1=\ ?$

$F=\dfrac{\pi r^2Y}{2L}x_1=\dfrac{4\pi r^2Y}{L}x_2$

$x_2=\dfrac{x_1}{8}$

$\dfrac{9x_1}{8}=\dfrac{27}{4}\Rightarrow x_1=6\ \text{mm}$

25. $\delta=10^4\ \text{kg/m}^3$

$\dfrac{T}{A}=\sigma$ ____ (1) from (1) & (3)

$\sigma=\delta\times10^8\ N/m^2$

$m=4\pi gm$

$T=900\pi N$

$f=\dfrac{v}{2L}=\dfrac{1}{2L}\sqrt{\dfrac{T}{\mu}}$

$=\dfrac{T\delta}{2m\sigma}\sqrt{\dfrac{\sigma}{\delta}}$

$=\dfrac{T\delta}{2m}\sqrt{\dfrac{\sigma}{\delta}}$

$=\dfrac{900\pi}{2\times4\pi\times10^{-3}}\sqrt{\dfrac{10^4}{9\times10^8}}$

$\dfrac{9000}{8\times3}=\dfrac{3000}{3}=375\ \text{Hz}$

26. $V=\dfrac{dx}{dt}; a=V\dfrac{dv}{dx}$

$E=\dfrac{1}{2}mv^2+2\times\dfrac{1}{2}\times\dfrac{3mR^2}{2}\times\dfrac{V^2}{R^2}+\dfrac{1}{2}Kx^2$

$=2mV^2+\dfrac{1}{2}Kx^2$

$\dfrac{dE}{dx}=0\Rightarrow4ma+Kx=0$

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$$a=-\left(\frac{K}{4m}\right)x$$

$$T=4\pi\sqrt{\frac{m}{K}};\beta=2$$

27. Applying work energy theorem $W_{\text{gas}}+W_{\text{atm}}=\Delta KE=-\frac{1}{4}mu^2$

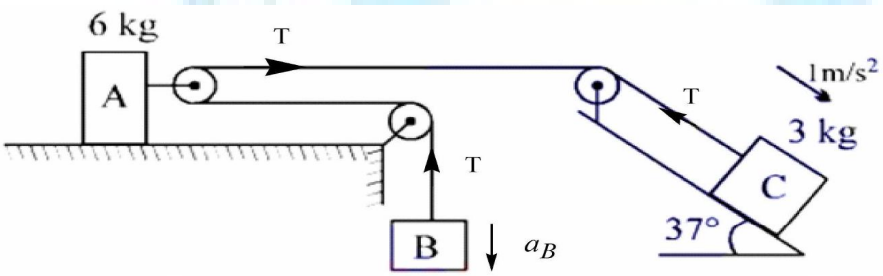
$$\Rightarrow W_{gas}=-\frac{1}{4}mu^2-W_{atm}=-\Delta U_{gas}$$

Since the gas is compressed

$$\therefore W_{\text{gas}}>0$$

$$\Rightarrow W_{\text{gas}}<-\frac{1}{4}mu^2 \text{ and } \Delta U_{\text{gas}}>\frac{1}{4}mu^2$$

28.



$$3\times10\times\frac{3}{5}-T=3\times1 \quad G_A=5\text{ m/s}^2$$

$$T=15N \quad a_B=9\text{ m/s}^2,\downarrow$$

$$mg-T=m\times g$$

$$m=15\text{ kg}$$

29. $F_1-F_2=(m_1+m_2)a_0$

$$kx-F_2=m_2a_0$$

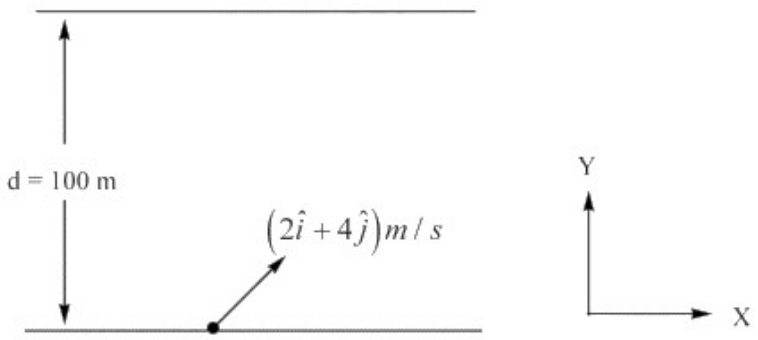
$$kx=m_2a_0+F_2$$

$$a_2=\frac{kx}{m_2}=a_0+\frac{F_2}{m_2}$$

$$a_1=a_0$$

$$\text{d}=100\text{ m}$$

30.



$$\vec{u}_b=\vec{u}_{b\to r}+\vec{u}_{r\to e}$$

$$u_b=2\sqrt{5}\text{ m/s}$$

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$= -2\hat{i} + 4\hat{j} + 4\hat{i}$

$= \left(2\hat{i} + 4\hat{j}\right) \text{m/s}$

$d = 100m$

$T = \frac{100}{4} = 25s$

drift $= 25 \times 2 = 50$ m

31. $\mu mg = 10N$

$20N = m \times 10m/s^2$

$m = 2kg, \mu = 0.5$

32. Beat frequency $= |f_1 - f_2|$.

33. $m = 2kg, R = 11N/m$

A: $R_{eq} = 200$ N/m

$a = 2m/s^2$

B: $a = \frac{1(3+1)}{2} = 2 \text{ m/s}^2$

C: $a = \frac{1}{2}(6+2) = 4m/s^2$

D: $a = \frac{1}{2}(5-3) = m/s^2$

34. Kinetic energy in com frame is always zero.

35. $\text{A}: I = \frac{Ml^2}{3}\sin^2\theta = \frac{MR}{12}$

$\text{B}: MR^2 = I_{cm} + M\left(\frac{2R}{\pi}\right)^2$

$I = I_{cm} + M\left(R - \frac{2R}{\pi}\right)^2 \#(1)$

$I = MR^2 + MR^2\left\{\left(1 - \frac{2}{\pi}\right)^2 - \left(\frac{2}{\pi}\right)^2\right\} \#(2)$

$= MR^2\left[1 + 1 - \frac{4}{\pi}\right]$

$= 2MR^2\left(1 - \frac{7}{\pi}\right) = \frac{SMR^2}{11}$

C: $I = \frac{1}{6}M\left(\frac{R\sqrt{3}}{2}\right)^2 = \frac{MR^2}{8}$

D: $i = \frac{M(2R)^2}{2} - \frac{M}{4}\left\{\frac{R^2}{2} + R^2\right\} = MR^2\left[2 - \frac{3}{8}\right]$

$= \frac{13MR^2}{8}$

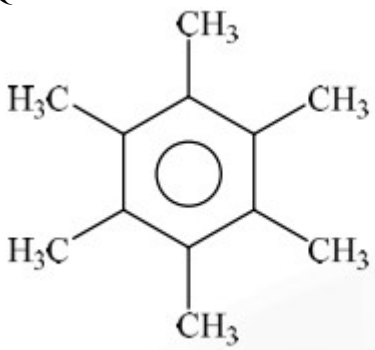
36. For siphon to work: $h_1 + h_2 \le \frac{P}{\rho g}$

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CHEMISTRY

37. $P: CH_3-C\equiv C-CH_3$

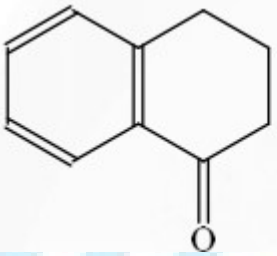
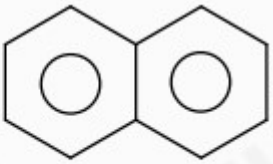
Q:



$(C_{12}H_{18})$

$$w\%\ \text{of}\ C=\frac{144}{162}\times 100=88.89\%$$

38.

X is  , Y is  $\equiv C_{10}H_8$

39. $\text{Vlt}\ 1MS+\text{Vlt}1M\ (S,R)$

$$V\text{mol}S+\frac{V}{2}\text{mol}S+\frac{V}{2}\text{mol}R$$

$$[\alpha]=\frac{3V}{2\times 2V}\times(+13)+\frac{V}{2\times 2V}\times(-13)=+6.5^{\circ}$$

$$+6.5=\frac{\alpha\times 1000}{1\times 88}\Rightarrow \alpha=6.5\times 88/1000=0.57$$

40. $X=120\left(sp^2\text{ - hybridization }\right)$

$Y=180(sp-\text{ hybridization }\ )$

$Z=90\left(sp^3d^2-\text{ hybridization }\right)$

$$\frac{120+180+90}{10}=\frac{390}{10}=39$$

41. In trigonal bipyramidal angels are 9 and square pyramidal angels are 8

42.

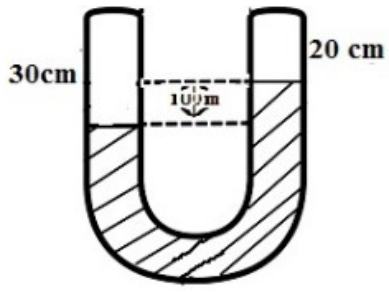
	$NH_3(g)$	$HCl(g)\rightarrow$	$NH_4Cl(s)$
Ini moles	$\frac{5\times 3}{RT}$	$\frac{2\times 4}{RT}$	-
Final moles	$\frac{7}{RT}$	-	$\frac{8}{RT}$

$$P_fV_f=n_fRT\Rightarrow P_f\times 7=\frac{7}{RT}\times RT$$

$$\Rightarrow P_f=1\ \text{atm}$$

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43.



$$P_{\text{right}} = \frac{80\times 25}{20} = 100 \text{ cm of Hg}$$

$$P_{\text{left}} = 100+10 = 110 \text{ cm of Hg}$$

$$\alpha = \left[\frac{(PV)_{\text{final}}}{(PV)_{\text{inital}}} - 1\right] \text{ for } NH_3$$

$$= \frac{110\times 30}{80\times 25} - 1 = 0.65$$

44. After addition of 0.2 MNaOH , since $pH = pK_a$, a buffer solution consisting of equal moles of HOCl and NaOCl has beed formed.

$$\left(pH = pK_a \frac{[\text{NaOCl}]}{[\text{HOCl}]}\right)$$

Let volume of original HOCl solution $= V mL$

$\text{HOCl} + \text{NaOH} \rightarrow \text{NaOCl} + \text{H}_2\text{O}$

1 mol of 0.2 V $(20\times 0.2) = 4$ -

Equilibrium $(0.2V - 4)$ - 4

$0.2V - 4 = 4$

$\Rightarrow V = 40\text{ml}$

After addition of 40 ml NaOH

$\text{HOCl} + \text{NaOH} \rightarrow \text{NaOCl} + \text{H}_2\text{O}$

mmol $0.2\times 40\ 0.2\times 40$

equilibrium - - 8

$$\therefore [\text{NaOCl}] = \frac{8}{40+40} = 0.1M$$

$$pH = \frac{1}{2}(pK_w + pK_a + \log C)$$

$$= \frac{1}{2}(14 + 7.5 - 1) = 10.25$$

45.

	Ka	$\Delta G^\circ C$(kJ / mole)	$\Delta H^\circ C$(kJ / mole)	$T\Delta S^\circ$(kJ / mole)
HCOOH	17.7×10^{-5}	+21	-0.4	-21.3 kJ
CH_3COOH	1.75×10^{-5}	+27.2	-0.5	-27.6 kJ
$\begin{matrix} CH_2-COOH \\ \vert \\ Cl \end{matrix}$	136×10^{-5}	+16	-4.6	-21 kJ

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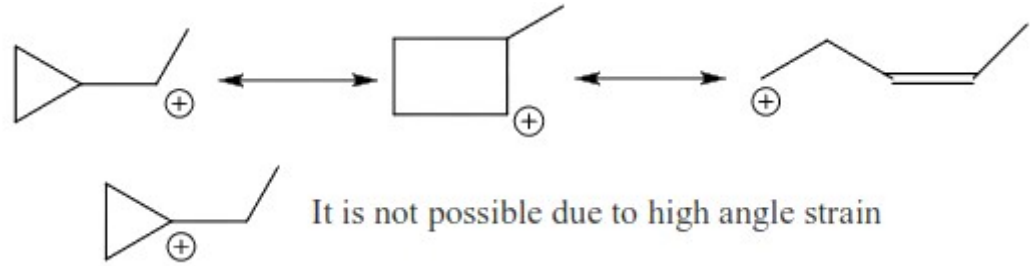
→ Ions formed in ionization are highly solvated, as a result ΔS° is $-ve$

→ From the above data, greater contribution for acidic strength is ΔS° (i.e solvation of ions only)

→ E.D.G increases the charge density at $-COO^-$ as a result, solvation of $-COO^-$ increases. Then ΔS° become more $-ve, \Delta G^\circ$ becomes more $+ve$. Finally acidic strength of carboxylic acid decreases.

→ E.W.G decreases the charge density at $-COO^-$ a sa result, salvation of $-COO^-$ decreases. Then ΔS°v = become less $-ve, \Delta G^\circ$ becomes less $+ve$. Finally acidic strength of carboxylic acid increases.

46.



It is not possible due to high angle strain

47. NO_2^- - bond order 1.5

NO_3^- - bond order 1.33

$\underset{113}{N_2O} < \underset{175}{N_2O_4} < \underset{186\ (\text{in pico meters})}{N_2O_3}$ (Order of $N-N$ bond length)

48. Carbon shows more tendency to form $p\pi - p\pi$ bonds where as silicon produces unstable $3p\pi - 3p\pi$ bonds

49. Since partial pressure of O_2 is constant, upon increasing volume, equilibrium shift in the forward direction

50. a) $\Delta V_{1-2} = O$ (siothermal process)

$\Delta V_{2-3} = W_{2-3}$ (adiabatic process)

b) $\Delta S_{1-2} + \Delta S_{2-3} = \Delta S_{1-3} \left(\because' S' \text{ is a state function }\right)$

But $\Delta S_{2-3} = 0$ (Reversible adiabatic process)

$\therefore \Delta S_{1-2} = \Delta S_{1-3}$

C) W is not a state function

D) $Q_{1-3} + W_{1-3} = \Delta U_{1-3} = \Delta U_{2-3} = W_{2-3}$

51. Flourination: Step - I is exothermic, Step - II is exothermic, Overall reaction is highly exothermic.
Chlorination: Step - I is slightly exothermic, Step - II is exothermic, overall reaction is exothermic.
Bromination: Step - I is endothermic, Step - II is exothermic, overall reaction is exothermic.
Iodination: Step - I is endothermic, Step - II is exothermic, overall reaction is endothermic.
Step - I is R.D.S, Step - II is fast

52. Boiling point (K) - B > Al > Ga > In > Tl

Ionic radius - $Sb^{3+} < Bi^{3+} < N^{3-} < P^{3-} < As^{3-}$

Atomic radius (pm) - $Tl > In > Al > Ga > B$

Density $\left(g/cm^3\right)$ at 298 K – $N < P < As < Sb < Bi < Pb$

53. $3d_{xy} \rightarrow 2$ nodal planes intersecting in a straight line

$4d_{x^2-y^2} \rightarrow 2$ nodal planes and a spherical node $\rightarrow 1$ straight line, 2 circles

$3d_{z^2} \rightarrow 2$ concial nodal surfaces $\rightarrow$ Point

$4d_{z^2} \rightarrow 2$ concial nodal surfaces and a spherical node $\rightarrow$ Point, 2 circles

54. $1,2 \rightarrow$ Polytropic process. $3 \rightarrow$ Reversible adiabatic process, $4 \rightarrow$ isothermal

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